

silQ: Learned Search for Analog Equalizer Design Implementation Status and Experimental Findings

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Contents

Abstract	1
DISCLAIMER: AI USAGE	1
1 Introduction	1
1.1 Objective and system output	1
1.2 Evaluation objectives	1
2 Problem Statement and Corrected Specification	1
2.1 Published requirement	1
2.2 Engineering acceptance beyond numerical compliance	1
3 Research Objectives and Design Rationale	2
3.1 Amortized search and target refinement	2
3.2 Acceptance criteria and signal quality	2
4 System Architecture	2
4.1 Evaluation Pipeline	2
4.2 The Guard Layer	3
4.3 Learned Global Search (PPO)	3
4.4 Constrained Local Refinement (G3.2)	3
4.5 Inference Modes	3
4.6 Web Dashboard	3
5 Experimental Methodology	4
5.1 Simulation and training configuration	4
5.2 Search benchmarks and protocol controls	4
5.3 PVT qualification	4
5.4 SNR and eye-opening evaluation	4
6 Experimental Results	4
6.1 Delivered circuit and PVT sign-off	4
6.2 SNR sensitivity of eye opening	5
6.3 Inference against budget-matched chance	5
6.4 Published-specification pass versus physical validity	6
6.5 Surrogate accuracy without trajectory leakage	6
6.6 Design-space non-uniformity and corpus checks	6
7 Development Timeline	6
8 Relation to the Submitted Abstract	7
9 Implementation Status	8
9.1 Completed Components	8
9.2 Partial Implementations	8
9.3 Verification Coverage	8
10 Limitations	8
10.1 Training and verification coverage	8
10.2 Physical implementation and observability	8
10.3 Scope of contribution	8
11 Conclusion	9
12 References	9

Abstract

This report presents silQ, a simulation-based framework for sizing a source-degenerated CTLE in SKY130 for a 5 Gb/s NRZ link. The framework combines PPO feasibility search, constrained local refinement, and guarded ngspice evaluation. In a 31-specification sweep, automatic inference solved 30 requests at a median wall time of 16.8 s, against a budget-matched chance count of 20.3. The delivered candidate passed all 45 PVT corners with a worst-corner boost error of 1.081 dB; it was the only fully passing candidate among 22 evaluated. Post-hoc SNR testing produced eye-opening pass rates of 0.833 at 20 dB, 0.333 at 15 dB, and 0.000 at 10 dB and below. These results characterize the generated designs and the eye metric, without establishing SNR-aware policy behavior. The output is a sized netlist with measured specifications and verification status. Area estimation is analytic and the DFE is post-processing.

DISCLAIMER: AI USAGE

AI tools were used to assist code implementation and report drafting. Responsibility for the architecture, experimental design, preregistered acceptance criteria, verification, and interpretation of results was retained by the authors. Invalid evaluations and unsupported interface claims were withdrawn following review. The authors understand and can defend the implementation and its reported measurements.

1 Introduction

1.1 Objective and system output

silQ was developed to automate sizing of a source-degenerated continuous-time linear equalizer (CTLE) for a 5 Gb/s NRZ link with a 2.5 GHz Nyquist frequency. The `eqrl` package in `nebula-analog-r1` accepts a requested boost and channel loss, generates a sized SKY130 netlist, evaluates its electrical specifications through ngspice, and attaches verification status to the output. The aim is to replace manual, iterative sizing and exhaustive parameter-space sweeps with a guarded, learned search procedure that produces a verified candidate directly from a target specification.

1.2 Evaluation objectives

The evaluation covers request-level search cost, physical validity, PVT qualification, and signal-to-noise ratio (SNR) sensitivity. PPO provides a feasible starting point, G3.2 supplies local target refinement, and the guard layer controls acceptance. The principal outputs are a verified candidate netlist, a dated comparison of inference modes, and a post-hoc assessment of eye opening under added noise. SNR results are reported separately from nominal boost accuracy because the frozen policy does not observe noise.

The inference-mode comparison is recorded in `docs/STATUS.md`; the delivered candidate is recorded in `results/delivered_circuit.json`. The implementation and test snapshot is dated 12 September 2026.

2 Problem Statement and Corrected Specification

2.1 Published requirement

The target problem statement specifies the following objective:

AI/ML for Analog Circuit Design. Demonstrate a fully automated framework for schematic design of analog circuits using Reinforcement Learning. Given circuit specifications, build an automated design flow that sizes devices using fewer search spaces (lowest design time) to reach near-optimal solutions, with zero human intervention. Use case: Equalizer circuit design (PCIe PHY). Should take significantly lower time than sweeping all MOS, R, C, L parameter space, with zero human intervention.

The published electrical requirements in `docs/PROBLEM.md` are listed below. Physical-validity checks were added to these limits.

2.2 Engineering acceptance beyond numerical compliance

The published PVT requirement defines the sign-off grid: 5 process conditions, 3 supply settings, and 3 temperatures, giving 45 corners.

The corrected acceptance contract retains the published limits and adds physical-validity gates. In particular, $\text{boost} = G_{\text{peak}} - G_{\text{DC}}$ does not itself prevent low-frequency attenuation. A 0 dB DC-gain floor is added

Requirement	Published specification
Link and Nyquist	PCIe Gen 2, 5.0 Gbps NRZ; 2.5 GHz Nyquist
HF peaking	3–12 dB boost, tunable over 1.25–2.5 GHz
Topology	1-stage CTLE with variable source-degeneration R_s , C_s , and 1-tap DFE
Linearity	HD3 < −30 dB at 100 MHz differential input
Input-referred noise	< 1.5 mV _{rms} over 10 MHz–5 GHz
Power and area	< 15 mW; < 0.05 mm ² in a 130 nm PDK
Eye opening	> 0.4 UI horizontal and > 100 mV vertical
PVT	TT, SS, FF, SF, FS; VDD ±5%; 0–125 °C

Published requirements; source: `docs/PROBLEM.md`.

as a topology-specific acceptance criterion. The role-specific operating-region and headroom checks in `src/eqrl/guards.py` catch failures that scalar performance limits miss.

Search efficiency is assessed through simulation budgets and request latency; global optimality is unmeasured. Supported requests run end to end without manual sizing, with target boost and channel loss controlling optimization.

3 Research Objectives and Design Rationale

3.1 Amortized search and target refinement

The original proposal treated circuit sizing as a simulation-driven learning problem. That learning is amortized across requests. The frozen checkpoint, `results/seq.clean40k.zip`, was trained for 40,960 simulator steps, and a k-nearest-neighbor surrogate was fitted on 374,455 usable AC records from an archive of over 414,000 SPICE runs.

For the frozen policy alone, a correlation of $r = 0.114$ was measured between requested and achieved boost. Feasibility discovery is therefore separated from requested-boost precision: PPO selects a feasible region and G3.2 performs constrained local refinement. The policy’s target correlation does not establish reliable retargeting without that refinement. Per-request online learning and uncertainty-driven exploration have not been demonstrated.

3.2 Acceptance criteria and signal quality

The electrical boost definition permits numerical compliance through low-frequency attenuation. A candidate with roughly −5 dB DC gain passed all eight published checks, motivating explicit operating-point and gain constraints (`docs/PASS_VS_VALID.md`; `src/eqrl/guards.py`). These checks distinguish target matching from physical acceptability.

Receiver qualification also requires sufficient eye opening after noise is introduced. Nominal boost, device validity, and a noiseless eye score do not establish that condition. The generated designs are consequently evaluated through the noisy eye metric, and PVT sign-off is used to assess process, voltage, and temperature variation. Pareto-style selection is available in `dashboard/`; physical multi-objective implementation is partial because area is analytic, the DFE is post-processing, and PVT is evaluated after optimization.

4 System Architecture

4.1 Evaluation Pipeline

The request enters through `src/eqrl/llm/spec_parser.py`; channel loss and requested boost are the operative inputs. A keyword fallback handles requests when no API key is present. PPO emits a continuous sizing action, `src/eqrl/circuits/ctle.py` generates the candidate netlist, and ngspice 47 evaluates real SKY130 device models.

Figure 1 illustrates the path from request to delivery. The guard rejects invalid candidates before they can influence reward or be returned. It verifies a refined design again before optional PVT sign-off. The netlist header records which checks were completed.

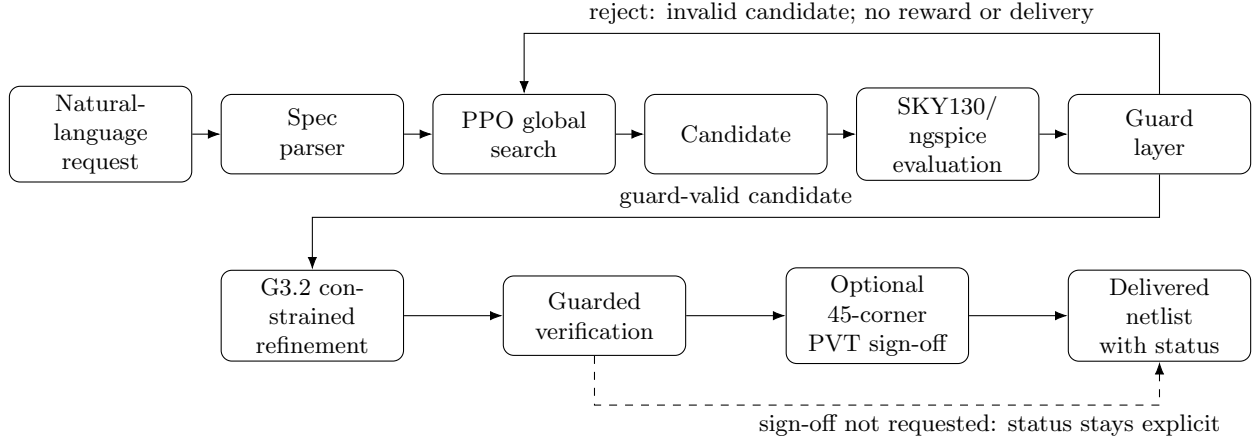


Figure 1: Guarded evaluation and delivery pipeline. The rejection loop prevents invalid candidates from contributing reward or being returned. Source: `src/eqrl/guards.py`.

4.2 The Guard Layer

The guard layer in `src/eqrl/guards.py` implements 20 checks in 5 tiers. It checks device operating region by role, bias and current-mirror sanity, convergence and finiteness, node headroom, and parameters pinned to a range edge. The output common-mode condition is $[V_{ov}, V_{DD} - V_{ov}]$. A favorable boost score cannot override these checks.

The measurement path is sealed: `measure_all` cannot be reached unguarded. Callers cannot bypass validation by forgetting a helper. The tiers cover run integrity, circuit sanity, corner integrity, physical plausibility, and search pathology; the last checks the experiment as a whole. The guard establishes conformance to these written criteria. It does not prove every aspect of functionality.

4.3 Learned Global Search (PPO)

PPO uses an 18-dimensional observation, a 6-dimensional continuous action, and a 20-step rollout. The frozen checkpoint is `results/seq_clean40k.zip`, at 40,960 training steps.

The reward function, observation vector, and guard thresholds were frozen on 26 August 2026. Subsequent changes were restricted to inference or separately named experimental arms.

4.4 Constrained Local Refinement (G3.2)

G3.2 refines requested boost around the PPO solution while respecting the acceptance constraints. The guard verifies each proposed refinement before delivery. The named `g32_acceptance` arm is evaluated separately from the five-mode comparison.

G3.2 is used for nominal target precision. The evaluation did not show improved corner robustness.

4.5 Inference Modes

The five inference modes are `default`, `fastest`, `thinking`, `retarget`, and `auto`. Table 3 reports their costs and solve counts from `docs/STATUS.md`.

The kNN surrogate reuses the development archive, with accuracy assessed on a chronological split. Its predictions do not replace SPICE verification or the guard checks.

4.6 Web Dashboard

The web application is served by `server.py` from `dashboard/`. The Design studio handles requests and alternatives, the Evidence page presents results, and the Lab exposes experiments. The interface reports verification status from the evaluated candidate.

The delivered netlist carries verification status in its header, so that status travels with the file. The full sign-off record, `results/pvt_signoff_seed23.json`, includes HD3 and noise simulation at every corner of the 45-corner grid. It verifies the generated candidate; training used only the typical corner.

5 Experimental Methodology

5.1 Simulation and training configuration

The circuit is evaluated with ngspice 47 and real SKY130 device models. A resident simulator loads the process models once and is re-primed between candidates. The measured evaluation time decreased from 6370 ms (about 6.37 s) to 77.5 ms, approximately $82\times$ (`results/speedup.json`); parallel environments separately improved the training-step bottleneck by approximately $241\times$. An earlier $350\times$ claim was replaced with these measurements.

The reward function, observation vector, and guard thresholds were frozen on 26 August 2026. All nine training configurations used the typical corner with `pvt: False`, a 5.0–11.0 dB target range, and the frozen policy in `results/seq-clean40k.zip`. PVT and SNR tests are performed as post-hoc evaluations.

5.2 Search benchmarks and protocol controls

Inference modes are compared on 31 requested specifications against a per-request, budget-matched chance baseline drawn from the achieved-design distribution without access to the target. The separate 32-spec benchmark uses targets from 5.01–10.99 dB and is in-distribution. Its results are not pooled with the 31-spec sweep.

An initial 32-of-32 result at a median of 1 simulation was excluded because each episode started from an already satisfying design with evaluation noise set to zero. After clean retraining, 26 of 32 was obtained at a median of roughly 4 evaluations, compared with classical-search counts of 16, 8, and 8 under the matched benchmark. These historical counts do not have a recorded matched chance line here. A separate target-weighted training comparison was confounded because enabling target tolerance also changed `passed`, which gated terminal reward.

5.3 PVT qualification

All 22 held-out candidates satisfying the strict typical-corner criterion were evaluated across the full 45-corner grid, using a selection rule specified before the corner runs. HD3 and input-referred noise were simulated at every corner, and each result was checked through the guard layer. Candidate-specific sign-off is recorded in `results/delivered_circuit.json`; the full verification path is also represented by `results/pvt_signoff_seed23.json`.

5.4 SNR and eye-opening evaluation

Eye-opening qualification is evaluated under added random noise. The vertical eye metric is the lowest scored logical-one level minus the highest scored logical-zero level:

$$E_{\text{eye}} = \min(v_{\text{one}}) - \max(v_{\text{zero}}).$$

Qualification requires a vertical opening greater than 100 mV. Because the metric uses extrema over the scored symbols, added noise can reduce the reported opening even when nominal gain and boost are unchanged. An approximate penalty of seven noise standard deviations is reported:

$$E_{\text{eye}} \approx E_{\text{eye},0} - 7\sigma_{\text{noise}}, \quad E_{\text{eye},0} \leq 2A,$$

where $E_{\text{eye},0}$ is the noiseless opening, σ_{noise} is the added-noise standard deviation, and A is the main signal amplitude. The approximation is specific to the studied scoring conditions.

The frozen policy uses a noiseless eye scorer and has no noise term in its observation vector. Accordingly, the SNR sweep tests output signal quality without retraining or noise-conditioned sizing. This test is distinct from the input-referred circuit-noise requirement of $1.5 \text{ mV}_{\text{rms}}$ over 10 MHz–5 GHz.

6 Experimental Results

6.1 Delivered circuit and PVT sign-off

The delivered CTLE targets 8.920 dB boost for a 14.83 dB channel. It passed the guard checks and all 45 PVT corners. Table 1 reports the output qualification recorded in `results/delivered_circuit.json`.

This candidate was the only fully passing design among the 22 qualified typical-corner candidates subjected to sign-off. The result establishes verification of the delivered output across the tested grid; it does not establish general PVT robustness of the generation policy.

Output characteristic	Reported result
Requested boost	8.920 dB
Channel loss	14.83 dB
Guard validity	All guard checks passed
PVT qualification	45 / 45 corners passed
Worst-corner boost error	1.081 dB
Worst-error corner	FS process; 1.71 V; 125 °C
Binding constraint	DC gain: 1.04 dB above the 0 dB floor

Table 1: Delivered circuit qualification. Source: `results/delivered_circuit.json`.

6.2 SNR sensitivity of eye opening

Table 2 reports the post-hoc SNR sweep. At 20 dB, the eye-opening pass rate was 0.833, equal to the noiseless reference rate. At 15 dB it was 0.333; at 10 dB and below, no evaluated design passed the eye-opening criterion. These rates characterize noisy signal qualification, independently of target-solving performance.

SNR	Pass rate	Observed qualification
20 dB	0.833	Equal to the noiseless reference rate
15 dB	0.333	Reduced eye-opening qualification
10 dB and below	0.000	No passing eye opening

Table 2: Eye-opening qualification under added noise. Source: post-hoc SNR sensitivity study.

Measured behavior. A loss of eye-opening qualification is observed as SNR decreases. The unchanged nominal boost, target error, valid-design count, and reward do not contradict this result: those quantities remain SNR-invariant because the frozen evaluation path does not include the added noise. Their invariance cannot be interpreted as evidence of noise tolerance.

Metric-dependent feasibility limit. Using the eye-opening bound in Section 5.4, the 100 mV criterion is estimated to become infeasible below approximately 11.8 dB SNR under the evaluated amplitude and noise assumptions. This is an approximate analytical limit, not an additional measured sweep point or a universal receiver threshold. It depends on the extrema-based eye scorer and the study conditions. SNR-aware retraining alone would not remove that metric-dependent limit.

Implication for the delivered output. The results support separate reporting of nominal electrical compliance, PVT qualification, and noisy eye opening. The 45-corner sign-off in Table 1 does not establish a per-SNR operating guarantee for the delivered netlist. SNR-aware optimization and such a candidate-specific operating guarantee have not been demonstrated.

6.3 Inference against budget-matched chance

Table 3 reports the 31-spec sweep. “Chance” is the expected solved-design count from specification-blind draws at the same simulation budget; “vs chance” gives the difference in counts.

Mode	Solved	Median wall	Sims (med/mean)	Chance	vs chance
default	18/31	86.8 s	14 / 16.0	29.8	−11.8
fastest	24/31	14.8 s	2 / 2.5	18.1	+5.9
thinking	30/31	127.9 s	32 / 40.7	30.5	−0.5
retarget	18/31	84.3 s	14 / 16.0	29.8	−11.8
auto	30/31	16.8 s	2 / 15.7	20.3	+9.7

Table 3: Inference modes against a budget-matched chance baseline. Measured 07 Sep 2026; source: `docs/STATUS.md`, 31-spec sweep.

At 40 simulations, specification-blind drawing already reaches 30.5, so the 30/31 result for `thinking` sits at chance. The positive margins belong to `fastest` and `auto`; `default` and `retarget` finish roughly 12

designs below chance. Separately, the median of 1 simulation per new request directly answers the problem statement’s per-request search objective, excluding historical training cost.

6.4 Published-specification pass versus physical validity

Outcome	Reported result
Designs passing all eight published electrical specifications	28
Passing designs rejected by the full validity guard	24 (86%)
Passing designs failing the saturation criterion	36%

Table 4: Pass versus valid. Source: docs/PASS_VS_VALID.md.

Of 28 passing designs, 24 fail the full guard set: 36% fail the role-specific saturation criterion alone and 86% fail the full set. The saturation failures hold independently of the additional DC-gain floor.

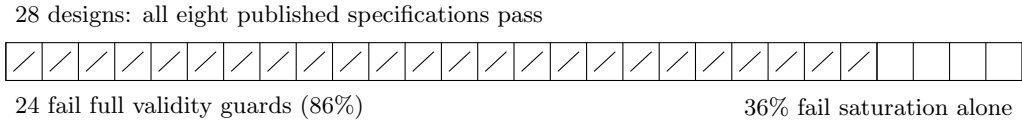


Figure 2: Each segment is one passing design; a diagonal stroke marks rejection by the full guard set. Saturation failures are included in that set. Source: docs/PASS_VS_VALID.md.

Figure 2 illustrates the mismatch. The result points to a wider problem in evaluating circuit optimizers:

Passing the published numerical benchmark $\neq$ producing a valid engineering design.

6.5 Surrogate accuracy without trajectory leakage

Boost-prediction evaluation	MAE	R^2
kNN; chronological split	≈ 0.295 dB	≈ 0.839

Table 5: Surrogate model accuracy. Source: chronological evaluation of the 374,455-record AC corpus.

An initial split of the 374,455-record corpus placed adjacent PPO trajectory points in both training and test. This leakage was removed through chronological, era-based partitioning, and the resulting accuracy is reported. The development archive supplied the data needed for surrogate fitting, which had initially been excluded for lack of data.

6.6 Design-space non-uniformity and corpus checks

Distinct guard-valid draw within ± 1.5 dB	Probability
Average over requested targets	0.232
At an 11 dB request	0.086

Table 6: Design-space non-uniformity. Source: distinct guard-valid achieved-design corpus.

More high-boost requests can make an equally capable method look worse; results should be reported by target region. Separately, the 0 dB DC-gain floor leaves 251,391 of 374,588 designs, with maximum boost 19.17 dB: it does not cut off the required 3–12 dB range, but these figures do not establish uniform coverage. Approximately 4% are clean across all 45 PVT corners; this corpus result does not guarantee robust future outputs.

7 Development Timeline

The project ran from 14 August to 12 September 2026 across 133 commits. The sequence below is reconstructed from that history and from the dated artifacts it references; each entry names the measurement that closed it.

Date	Recorded event and outcome
14 Aug	Scaffold; source-degenerated CTLE evaluated through resident ngspice with real SKY130 models. First end-to-end request-to-netlist loop.
15–16 Aug	Guard layer added and wired into the evaluator; all eight published specifications scored during training.
17 Aug	40,960 guarded steps end at 99.9% invalid. Search space measured; i_{tail} capped at 1 mA, raising the valid fraction from 10.4% to 20.4%. An earlier 45/45 claim re-measured on the design it referred to: 42/45.
18 Aug	Named baseline measured: 2,394 evaluations to first success. Policy solves 0 of 8 held-out specifications at 4,096 steps. Reward found not to score DC gain — the policy had learned to attenuate — and corrected. Vectorized environments, $\approx 241\times$ on the training-step bottleneck.
23 Aug	Matched-benchmark audit: PPO finds feasible circuits; no arm hits a requested target.
24 Aug	Week-1 protocol pre-registered before its compute ran. Surrogate arm cut when its corpus was found mislabeled.
26 Aug	G3.2 constrained refinement accepted on the held-out comparison: precision is real, solve count is not. Architecture frozen — reward, observation vector and guard thresholds.
27 Aug	45-corner sign-off over 22 candidates: one survives. Delivered circuit frozen.
30–31 Aug	Target-conditioned reward arm: not supported. G3.2 slope constants measured to be runtime-derivable rather than hand-tuned.
04–06 Sep	DFE tap implemented; parser restricted to fields it actually reads; five inference modes built; retarget measured at 0/32 on a fresh seed.
12 Sep	Verification snapshot: 908 passing Python tests, 3 skipped, 2 xfailed, 50 Node tests.

Table 7: Development timeline, 14 August–12 September 2026. Source: commit history and the dated artifacts each entry cites.

Five entries record a measurement withdrawing an earlier claim of this project: the 45/45 result re-measured as 42/45, the reward that did not score DC gain, the excluded 32-of-32 benchmark, the surrogate arm cut on a corpus defect, and the $350\times$ speedup replaced by $82\times$. Those corrections are the reason the numbers reported here carry their measurement dates and artifact paths.

8 Relation to the Submitted Abstract

The submitted abstract proposed an agent that “learns the circuit while designing it”, building its own model of the design space from every simulation, deciding “where to explore, what to test and which designs are worth pursuing”, and maintaining multiple high-performing tradeoffs across gain, bandwidth, noise, power and area. Four divergences from that proposal are recorded below with the measurement that produced each.

Per-session learning became amortized learning. The abstract’s loop rebuilds its model on every request. The delivered system learns once — 40,960 PPO steps through the real simulator, plus a corpus of 374,588 recorded runs — and reuses that knowledge, answering a request at a median of 2 simulations in **fastest** and **auto** (Table 3). Measured against the abstract’s own objective of reducing unnecessary simulations, and against the problem statement’s “fewer search spaces (lowest design time)”, amortization is the stronger implementation: knowledge persists across requests rather than expiring with a session. What is not delivered is the uncertainty-driven exploration the abstract describes; the system does not choose its next experiment from a posterior.

One agent became an explicit division of labour. The abstract assumes a single agent that understands the space. Measurement gives $r = 0.114$ for requested versus achieved boost: the policy had learned feasibility, not intent. Responsibility was therefore split — global search for the feasible basin, constrained local refinement for precision — rather than claiming an understanding the data does not support.

Smarter search became a trustworthy objective. The abstract assumes the objective is sound and the difficulty is search efficiency. Section 6.4 shows the reverse: the published boost definition admits a -5 dB DC-gain solution that satisfies all eight published specifications. The highest-value work moved from the searcher to the evaluator, and the guard layer, built as protection against reward hacking, became the instrument that exposed the gap.

Multi-objective tradeoffs are partially delivered. Pareto-style selection over competing objectives exists in the interface. However, PVT is post-hoc verification rather than part of the policy state, area

is computed analytically rather than extracted from layout, and the DFE is post-processing rather than a transistor-level implementation. This is a partial delivery of the abstract’s promise and is labelled as such.

One addition the abstract did not anticipate. Every solve rate reported here carries a budget-matched, specification-blind chance line. No such baseline was found in the 16 RL-for-analog-sizing papers surveyed on 07 September 2026. Table 3 shows it changing a conclusion rather than annotating one: read as a ranking of solve counts, 30/31 is the best result; read against chance, it is the mode that sits on its own baseline.

9 Implementation Status

9.1 Completed Components

The `eqrl` implementation connects specification parsing, parametric CTLE generation, PPO inference, G3.2 refinement, ngspice evaluation, and guarded delivery. The output is a sized netlist with resulting specifications and a verification header. The Design, Evidence, and Lab views are served by `server.py` from `dashboard/`; the parser includes a keyword fallback in `src/eqrl/llm/spec_parser.py`.

9.2 Partial Implementations

The parser recognizes roughly 13 fields, but only target boost and channel loss control optimization. Unsupported requirements are refused. Pareto-style alternatives are implemented; area is analytic and the DFE is post-processing. PVT and SNR are evaluated after nominal optimization. Neither corner variation nor added noise is part of the frozen policy state.

9.3 Verification Coverage

As of 12 September 2026, the test suite reports 908 passing Python tests, with 3 skipped and 2 xfailed, plus 50 Node tests across five files. Regression tests cover known reward exploits, and ratchet tests compare documented numerical claims against the cited code. Reproduction instructions are in `docs/REPRODUCE.md`. Electrical qualification is supported by the simulator records; software tests do not substitute for those measurements.

10 Limitations

10.1 Training and verification coverage

Training covers 5.0–11.0 dB at the typical corner, while the published requirement is 3–12 dB. Generalization outside the training band has not been measured. The 32-spec benchmark is in-distribution by construction. G3.2 supplies nominal target refinement but did not improve corner robustness; only 1 of the 22 candidates in the delivered-circuit selection passed all 45 corners.

The SNR sweep is post-hoc. The frozen policy has no noise-conditioned input, and nominal score invariance does not demonstrate SNR robustness. The approximately 11.8 dB limit applies to the evaluated eye metric and its assumptions. The aggregate SNR pass rates do not establish a noise limit for the delivered candidate.

10.2 Physical implementation and observability

Area has not been extracted from layout, nor has the DFE been implemented at transistor level. No physical equalizer, FPGA, STM32 implementation, or sim-to-real transfer is provided. The 0 dB DC-gain floor is an implementation-specific acceptance criterion.

The training runs produced no reward curves because the vectorized environment was constructed without a monitor wrapper. An exhaustive-sweep wall-time comparison and an RL-versus-search sample-efficiency curve have not been measured. The reported median of 1 simulation concerns amortized request cost and excludes training expenditure.

10.3 Scope of contribution

RL-based analog sizing precedes this project, including AutoCkt (DATE 2020), DNN-Opt (2021), and PVT-aware and goal-conditioned approaches published in 2025. The contribution of this project is the evaluated integration of learned feasibility search, constrained refinement, a reward-integrity guard, and budget-matched chance comparison. No budget-matched, specification-blind baseline was identified among the 16 surveyed RL-for-analog-sizing papers; that survey does not establish priority over the field.

11 Conclusion

This project produced a guarded CTLE netlist with silQ that passed all 45 tested PVT corners with a worst-corner boost error of 1.081 dB. Automatic inference solved 30 of 31 requested specifications at a median wall time of 16.8 s, against a budget-matched chance count of 20.3. The delivered candidate was the only fully passing output among 22 candidates subjected to corner sign-off.

The SNR sweep exposed a separate signal-quality limitation: eye-opening qualification decreased from 0.833 at 20 dB to 0.333 at 15 dB and 0.000 at 10 dB and below. The approximate 11.8 dB feasibility limit is specific to the evaluated eye metric. The demonstrated output is a nominally optimized, post-hoc-qualified circuit; SNR-aware optimization, broader corner robustness, and physical implementation require further evaluation.

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